

Prof. Barry Quinn

Professor of Finance and Financial Technology

Contact Information

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- **Date of Birth:** 10 July 1973

Personal Statement

Professor Barry Quinn is Professor of Finance & Financial Technology at Ulster University Business School and Director of the Centre for Finance and Responsible Technology. The Centre is guided by three pillars: statistical rigour and intellectual humility; practical industry relevance; and responsible algorithmic innovation.

Barry is a Chartered Statistician and views econometrics as an applied statistical science. His work combines applied econometrics and causal inference with trustworthy machine learning and computational approaches to produce decision-relevant evidence for markets, risk, market integrity, and regulatory compliance. In practice, that means making assumptions explicit, quantifying uncertainty, stress-testing results, and being clear about limits—so claims remain proportionate to evidence.

His research spans causal policy evaluation, algorithmic trading risk detection, and trustworthy AI methods. He works collaboratively across finance, statistics, computer science, and law, and teaches across quantitative finance, econometrics, and AI applications in finance, emphasising ethical practice, reproducibility, and critical thinking.

Before entering academia, Barry worked in financial markets, specialising in currency trading and liquidity management. He holds a PhD in Finance and an MSc in Artificial Intelligence from Queen's University Belfast, and works with external partners to ensure research translates into robust practice.

Education and Professional Qualifications

Qualification	Institution	Year
B.Sc.(Hons) Accounting and Finance	Queen’s University Belfast	1995
MSc Quantitative Finance	RMIT University Melbourne	2006
Ph.D. Finance	Queen’s University Belfast	2012
Chartered Statistician	Royal Statistical Society	2019
Advanced Data Science Professional	Royal Statistical Society	2023
MSc Artificial Intelligence	Queen’s University Belfast	2025

Research Excellence

Research Programme

My research develops across three methodological strands.

The first is **trustworthy machine learning for market integrity and risk**. This includes representation-learning approaches to market-manipulation detection, including work published in *Information Processing & Management*, and related projects on anomaly detection, signal amplification, and sentiment-conditioned crash-risk modelling. The methodological commitment throughout is that model performance and model trustworthiness are separable properties, both of which must be evidenced before deployment in operational or regulatory settings.

The second is **computer vision for socio-economic measurement**. This work uses real-world imagery to develop auditable indicators of social, economic, and regulatory phenomena, including projects on environmental correlates of vehicle crime and the measurement of cultural-symbol displays in Northern Ireland. The aim is to treat vision pipelines as measurement instruments rather than accuracy-maximising systems alone.

The third is **causal and statistical evidence for governance and regulation**. This includes work on innovation signals and strategic exits in RegTech, Basel III’s cross-border transmission through foreign bank subsidiaries, credit-union risk and resilience, and collaborative doctoral partnerships with industry partners including Pytillia, Napier AI, and Funds Axis. Across all three strands, the priority is that evidence survive operational and regulatory scrutiny: assumptions explicit, uncertainty quantified, and claims proportionate to what the data can support.

Peer-Reviewed Publications

- Dai, Yongsheng, Quinn, Barry, Kearney, Fearghal, Liu, Weilong, Spence, Ivor, Rafferty, Karen, Wang, Hui (2026). Detecting market manipulation with dual-branch self-supervised learning: A unified framework integrating frequency-informed anomaly synthesis and domain-specific features. *Information Processing & Management*, 63(8), Article 104961, 1–30.
- Wang, Hui, Dai, Yongsheng, Spence, Ivor, Rafferty, Karen, Quinn, Barry, Huang, Ji (2025). TDSRL: Time Series Dual Self-Supervised Representation Learning for Anomaly Detection from Different Perspectives. *IEEE Internet of Things Journal*.
- Peng, Qiao (Olivia), McKillop, Donal, Quinn, Barry, Liu, Kailong (2025). Modeling and predicting failure in US credit unions. *International Journal of Forecasting*, 41(3), 1237–1259.
- Liu, Weilong, Zhang, Yong, Liu, Kailong, Quinn, Barry, Yang, Xingyu, Peng, Qiao (2024). Evolutionary Multi-Objective Optimisation for Large-Scale Portfolio Selection With Both Random and Uncertain Returns. *IEEE Transactions on Evolutionary Computation*.

- Bouri, E., Quinn, B., Sheenan, L. & Tang, Y. (2024). Investigating extreme linkage topology in the aerospace and defence industry. *International Review of Financial Analysis*.
- Quinn, Barry, Gallagher, Ronan, Kuosmanen, Timo (2023). Lurking in the shadows: The impact of CO2 emissions target setting on carbon pricing in the Kyoto agreement period. *Energy Economics*.
- Liu, Jiadong, Papailias, Fotis, Quinn, Barry (2021). Direction-of-change forecasting in commodity futures markets. *International Review of Financial Analysis*.
- Gallagher, Ronan, Quinn, Barry (2020). Regulatory own goals: The unintended consequences of economic regulation in professional football. *European Sport Management Quarterly*.
- McKillop, Donal, French, Declan, Quinn, Barry, Sobiech, Anna L, Wilson, John OS (2020). Cooperative financial institutions: A review of the literature. *International Review of Financial Analysis*.
- Quinn, Barry, Hanna, Alan, MacDonald, Fred (2018). Picking up the pennies in front of the bulldozer: The profitability of gilt based trading strategies. *Finance Research Letters*.
- McKillop, Donal G, Quinn, Barry (2017). Irish credit unions: Differential regulation based on business model complexity. *The British Accounting Review*.
- Ayadi, Rym, Naceur, Sami Ben, Casu, Barbara, Quinn, Barry (2016). Does Basel compliance matter for bank performance? *Journal of Financial Stability*.
- Glass, J, McKillop, Donal, Quinn, Barry (2015). Modelling the Performance of Irish Credit Unions, 2002-2010. *Financial Accountability & Management*.
- McKillop, Donal G, Quinn, Barry (2015). Web adoption by Irish credit unions: Performance implications. *Annals of Public and Cooperative Economics*.
- Glass, J Colin, McKillop, Donal G, Quinn, Barry, Wilson, John (2014). Cooperative bank efficiency in Japan: a parametric distance function analysis. *The European Journal of Finance*.
- Quinn, Barry, McKillop, Donal (2009). Internet banking and Irish credit unions. *International Journal of Cooperative Management*.
- Quinn, Barry, McKillop, Donal (2009). Cost performance of Irish credit unions. *Journal of Cooperative Studies*.

Research Impact and Funding

Table 1: Major Research Grants (Selected)

Funder	Role	Years	Project	Costs
DfE NI	PI	2025	Three PhD scholarships to support the Centre for Finance and Responsible Technology (including two collaborative doctoral partnerships)	£360,000
Innovate UK	PI	2022-2023	E.S.G fair value analytics platform: using state-of-the-art financial data science and business analytics to design a fair-value ESG prediction engine	£173,000
Innovate UK	PI	2021-2023	Regulatory technology and portfolio analytics using state-of-the-art econometrics and financial machine learning	£173,000
Innovate UK	PI	2018-2021	Designing and deploying a retail analytics platform using advanced analytics and machine learning	£165,000
UKFin+	PI	2024-2025	Leveraging AI to understand and improve regulatory compliance in the Investment Management Industry	£100,000

Working Papers (Selected)

- Dai, Yongsheng, Quinn, Barry, Kearney, Fearghal, Wang, Hui. *Signal Amplification and Strategic Deterrence in Market Surveillance*. Working paper.
- Dai, Yongsheng, Huang, Ji, Ren, Spence, Ivor, Rafferty, Karen, Quinn, Barry, Wang, Hui. *FMISD: Fine-grained Multi-Scale Modeling for Inexact Supervised Anomaly Detection of Time Series*. Under revision at *IEEE Transactions on Cybernetics*.
- Quinn, Barry, Sheenan, Lisa, Zhang, Ying (Veronica). *Regulatory Reach or Regulatory Arbitrage? Basel III's Cross-Border Transmission Through Foreign Bank Subsidiaries*. Centre for Responsible Banking & Finance Working Paper 25-017.
- Hannon, James (Corresponding Author), French, Declan, Quinn, Barry, O'Hagan, Adrian. *Geospatial modeling of vehicle crime in Northern Ireland using computer vision to identify environmental factors*. Under revision.
- Cochrane, Brandon, Bryan, Dominic, Quinn, Barry, Graham, Bryon, French, Declan (2026). *Flying the Flag: Cultural Expression as Territorial Demarcation*. IGPSJ Working Paper WP 02/26, Queen's University Belfast.

PhD Supervision Excellence

Completed Supervisions (6):

1. Dr Jiadong Liu (graduated 2018): *Momentum in Empirical Asset Pricing* (**Principal Supervisor**).
2. Dr Ashleigh Neil (graduated 2019): *Law and Financial Stability* (**Principal Supervisor**).
3. Dr Colm Kelly (graduated 2021): *Machine Learning in Empirical Asset Pricing* (**Principal Supervisor**).
4. Veronica Zhang (2024): *Capital Policy and State Sponsorship in Chinese Banking* (**Principal Supervisor**).
5. Dr Kevin Johnson (graduated 2019): *A Case Study in Industrial Consolidation — The Irish Credit Union Sector* (**Co-Supervisor**).

6. Dr Qiao (Olivia) Peng (graduated 2021): *US Credit Union Mergers — Causes and Consequences* (**Co-Supervisor**).

Current Supervisions (6):

1. Fei Li: *Sustainable Investing, Reporting and Gender Diversity* (fully-funded QBS scholarship) (**Principal Supervisor**).
2. Abhishek Pramanick: *AI and the Future of Financial Regulation* (fully-funded QBS scholarship) (**Principal Supervisor**).
3. Dongwei (Jeffrey) Li: *The Economics of AI Patents and Productivity in Finance* (fully-funded DofE scholarship) (**Principal Supervisor**).
4. Yuqi Ding: *The Economics of Generative AI in Finance* (fully-funded QBS scholarship) (**Principal Supervisor**).
5. Yongsheng Dai: *Market Manipulation and Financial Anomaly Detection* (fully-funded PWC scholarship) (**Co-Supervisor**).
6. Brandon Cochrane: *Economic Cost of Cultural Displays in Northern Ireland* (fully-funded DofE CAST scholarship) (**Co-Supervisor**).

Teaching and Academic Leadership

Recent Teaching Innovations (Ulster University)

- **Programme Lead**, MiniMBA in FinTech and Responsible Data Science, Magee Campus (DfE-sponsored; inaugural cohort 2026; Compare NI FinTech Scholarship winner; next delivery Spring 2027).
- **Founder**, Trading and Investment Group (£100K alumni philanthropic fund; in partnership with Innovation Ulster Ltd; launching September 2026).
- **Open textbook**, *Statistical Science for Finance: Rigorous Methods for Technology-Enabled Markets* — companion resource for FIN306 FinTech and Data Science.
- **Teaching software**, [TickLab](https://ticklab.co.uk) — web-based trading simulation for market microstructure and classroom trading dynamics.
- **Current modules**: Financial Data Science; Causal AI; AI in Trading; Financial Econometrics.

Teaching Publications

- Quinn, Barry (2026). *Statistical Science for Finance: Rigorous Methods for Technology-Enabled Markets*. Open educational resource and companion textbook for FIN306, Ulster University Business School. <https://quinfer.github.io/financial-data-science/>
- Quinn, Barry (2026). *TickLab: Web-based trading simulation for market microstructure and trading dynamics*. Open teaching software for classroom use, with interactive tutorials, an online simulation engine, and academic foundations. www.ticklab.co.uk
- Quinn, Barry, Hanna, Alan, Gallagher, Aine (2018). Queen's Student Managed Fund: Investing in the Student Experience. *Reflections*, 27, pp. 16–17.

Earlier Teaching Leadership (Queen's University Belfast)

- **Programme Director**, MSc Quantitative Finance (2018–2022).
- **Programme Director**, MSc Computational Finance & Trading (2014–2018).

- **Co-founder**, Queen’s Student Managed Fund (2012–2024).
- **Trading Principles**: postgraduate market-microstructure course built around live in-class trading simulations; extended subsequently as TickLab.
- **Teaching Award**, QUB (2016).

Academic Service

- **Director**, Centre for Finance and Responsible Technology, Ulster University Business School, September 2025–.
- **Founder**, Trading and Investment Group, Ulster University Business School, 2026– (£100K alumni philanthropic fund; in partnership with Innovation Ulster Ltd; launching September 2026).
- **Programme Lead**, MiniMBA in FinTech and Responsible Data Science, Magee Campus, Ulster University Business School, 2026– (DfE-sponsored; credit-bearing four-day intensive (one day online); inaugural cohort 2026; Compare NI FinTech Scholarship winner from cohort; open materials at quinfer.github.io/minimba; next delivery Spring 2027).
- **Co-founder**, Finance and AI Research Lab (with Dr Fearghal Kearney), Queen’s University Belfast, 2022–2024.
- **Lead Developer**, Queen’s Business School Remote Analytics Lab Platform (Q-RaP), 2021–2024.
- **QUB Academic Lead**, Steering Group for Northern Ireland Global Centre in Secure Connected Intelligence for Regulatory Technology in Finance, 2022–2024.
- **Programme Director**, MSc Quantitative Finance, Queen’s University Belfast, 2018–2022.
- **Senior Research Fellow** (Sabbatical), International Research Centre on Cooperative Finance, HEC Montreal University, Canada, May–September 2018.
- **Programme Director**, MSc Computational Finance & Trading, Queen’s University Belfast, 2014–2018.
- **Co-founder**, Queen’s Student Managed Fund, 2012–2024.
- **Implementation Team Member**, Bloomberg Trading Room, Queen’s University Belfast.
- **External Grant Reviewer**, UKFin+ (UKRI specialist funded project) — reviewed feasibility grant application (£100K) for financial services innovation research, 2024.
- **Organiser**, Symposium on AI and the Future of Financial Regulation, Queen’s University Belfast, November 2023.
- **Organiser**, Symposium on Safety and Assurance in Finance, Queen’s Business School, November 2024.
- **Chair**, Panel Debate on “Digital Regulation: Shaping Digital Markets and Safeguarding Consumer Rights in Northern Ireland”, Northern Ireland Competition Forum, May 2024.

Professional Experience

Period	Position	Institution
2024 - Present	Professor of Finance & Financial Technology	Ulster University Business School
Sep 2025 - Present	Director, Centre for Finance and Responsible Technology	Ulster University Business School
2020 - 2024	Senior Lecturer in Finance, Technology and Data Science	Queen's Management School
2010 - 2020	Lecturer	Department of Finance, Queen's Management School
1998 - 2004	Currency Trader and Liquidity Manager	Janus Henderson Investors, London

Professional Recognition

- Chartered Statistician, Royal Statistical Society (2019).
- Advanced Data Science Professional, Royal Statistical Society (2023).
- **1st Place**, CFA European Quantitative Finance Awards (2018).
- **Associate Research Fellow**, QUB Momentum One Zero (2021–2024).